

On-line coupling of liquid chromatography at critical conditions with electrospray ionization tandem mass spectrometry for the characterization of a nitroxide-mediated poly(ethylene oxide)/polystyrene block copolymer

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Coupling of liquid chromatography at critical conditions (LCCC) with on-line mass spectrometry (MS) detection was implemented via an electrospray ionization (ESI) interface, using a mobile phase containing the cationizing agent. Critical conditions established for poly(ethylene oxide) were used to characterize a poly(ethylene oxide)/polystyrene block copolymer (PEO-*b*-PS) in both MS and MS/MS modes. As co-oligomer molecules were successfully separated according to the PS block size, structural information could be reached from simplified MS spectra. The microstructure of this copolymer, synthesized by nitroxide-mediated polymerization, could further be unambiguously characterized in LCCC/ESI-MS/MS experiments since the PS block size could be reached by both the co-oligomer chromatographic behavior and its MS/MS pattern. Copyright © 2008 John Wiley & Sons, Ltd.

Because of their intrinsic nature, polymers are complex materials. The synthesis of functional polymers and block, random or grafted copolymers generates a series of distribution owing to the presence, in the whole sample, of polymer chains with different functionalities, chemical compositions, architectures, etc. As a result, polymer characterization can be a tedious task if the polymers are analyzed as a mixture, without previous separation. Size-exclusion chromatography (SEC) is widely used for the determination of polymer molar mass distribution.^{1,2} However, this technique is poorly efficient because two species of different molar mass and different chemical compositions may have the same hydrodynamic volume. Moreover, weak adsorption phenomena could affect the polymer mass values determined by SEC. As a result, new chromatographic techniques have been developed for polymer separation, amongst which liquid chromatography at critical conditions (LCCC) is very promising. Pioneering studies were reported at the end of the 1980s by Gorbunov *et al.*³ and later by Berek *et al.*⁴ In the chromatographic separation of molecules using a porous stationary phase,

retention is determined by both size exclusion and interaction mechanisms. At the critical conditions, effects due to these two mechanisms compensate each other and the retention of homopolymer molecules becomes independent of their molecular weight.^{4–10} Chromatographic elution thus only depends on the chemical heterogeneity, oligomers of the same nature being eluted at the same retention time. This technique usually requires the use of binary eluent with a very precise composition and temperature. LCCC has attracted much interest for the characterization of functional homopolymers^{7,10–22} since retention is only determined by the end-groups. LCCC has also often been employed for the analysis of block copolymers^{23–31} since, by making one block chromatographically 'invisible', the retention time of co-oligomer molecules is mainly dictated by the size of the second block. The most important drawback of the LCCC technique is that the detection modes classically implemented (refractive index or ultraviolet detection) do not allow unambiguous identification of separated species.

Mass spectrometry (MS) has become an increasingly important tool for polymer analysis since the development of soft ionization methods.³² As oligomers can be detected as intact molecular adducts, data from matrix-assisted laser desorption/ionization (MALDI) or electrospray ionization (ESI) mass spectra can be utilized to determine the size of repeat units as well as end-group masses, average molecular weights and polymerization degree. Further structural characterization of the polymer repeat unit and end-groups

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requires oligomer ions to be fragmented, in collision-induced dissociation (CID) or post-source decay (PSD) experiments.³² However, in the particular case of copolymers, complex MS spectra are usually obtained since these molecules contain more than one type of repeat units. Moreover, when ESI is used for ionization, the presence of several charge states further complicates data analysis and increasing analyzer resolving power is required to efficiently resolve any peak overlap.³³

Therefore, the combination of LCCC with MS is expected to provide a powerful analytical tool since advantages offered by one technique would overcome the limitations inherent to the other one. The idea of combining LCCC with MS has already been reported.^{34–38} Most studies described an off-line approach, using LCCC separation in a preparative mode for further MS analysis. The so-collected fractions of polymer^{34,37,38} or copolymer³⁵ samples were then analyzed in a second MALDI-MS step. Nielen and Buijtenhuijs³⁶ developed a so-called 'pseudo-LCCC' separation method, which also allowed a mass-independent elution behaviour, with ESI-MS detection. Using a mixing Tee, post-column salt addition was performed to promote functionalized homopolymer ionization in the ESI source. As MS allows individual co-oligomer ions to be measured, the overall composition can be reached as long as some assumptions can be made regarding the mass of end-groups. However, in the case of unknown end-groups, the combination of LCCC and MS data might not always be sufficient to allow unambiguous characterization of the co-oligomer composition. In contrast, CID experiments can provide information on both the number and the position of repeat units throughout the molecule. Indeed, fragmentation pathways along the chain allow the copolymer to be sequenced since each co-monomer shows distinct dissociation behaviour.³² MS/MS data of copolymers can then be analyzed based on the knowledge of the fragmentation behaviour of homopolymers consisting of each of the co-monomers. Copolyethers, based on poly(ethylene oxide) (PEO) or poly(propylene oxide) (PPO)^{39–43} as well as copolyesters,^{33,44–47} have been the most studied synthetic copolymers in terms of fragmentation. A larger number of product ions are usually generated from copolymer than from homopolymer precursor ions.³² However, unexpectedly simple MS/MS spectra can be obtained, while still maintaining a high level of information, from macromolecules terminated with labile functionalities. As demonstrated for polystyrene (PS) homopolymers with a labile substituent at a chain end, a homolytic bond cleavage is induced upon CID which allows the elimination of the labile end-group as a radical and a subsequent depolymerization of the PS chain.^{48,49} We recently took advantage of such a depolymerization process to accurately determine the PS block size in a PEO-*b*-PS block copolymer from the number of styrene losses as observed in the MS/MS spectrum.⁵⁰ Alternative fragmentation pathways, involving selective dissociation of the function between the two blocks, were also shown to provide clear information about the block size.^{40,50}

In the present work, on-line coupling of LCCC with MS/MS has been implemented for the microstructural characterization of a PEO-*b*-PS block copolymer, using a mobile phase containing the electrospray cationizing agent.

EXPERIMENTAL

Chemicals

Methanol (MeOH) and HPLC grade water were purchased from SDS (Peypin, France) and used without further purification. Salt used to promote electrospray ionization was LiCl from Sigma-Aldrich (St. Louis, MO, USA), used at a 1 mM concentration in the mobile phase. Hydroxyl-terminated poly(ethylene oxide) standards, purchased from PSS (Mainz, Germany) and ranging from $M_w = 200$ to 10000 g mol^{-1} , were dissolved in MeOH and further diluted with the mobile phase to a final concentration of $100 \mu\text{g mL}^{-1}$.

All reaction steps for the preparation of SG1-capped PEO-*b*-PS (**3**) were performed under a dry nitrogen atmosphere. Poly(ethylene glycol) methyl ether (PEO-Me) with $M_w = 350 \text{ g mol}^{-1}$, styrene (99%), acryloyl chloride (96%) and triethylamine (TEA, 99%) were all purchased from Sigma-Aldrich and were used as received without further purification. The SG1-based alkoxyamine derived from methacrylic acid, also called MAMA-SG1 (Acid 2-methyl-2-[*N*-*tert*-butyl-*N*-(1-diethoxyphosphoryl-2,2-dimethyl propyl)aminoxyl]propanoic) was kindly supplied by Arkema (Lacq, France).

Synthesis of the SG1-capped PEO-*b*-PS copolymer

Synthesis of the SG1-capped PEO-*b*-PS copolymer has been described in detail elsewhere⁵⁰ and will only be briefly reported here. The synthetic procedure of copolymer **3** involves three steps: the preparation of PEO-acrylate (**1**), the intermolecular radical addition of MAMA-SG1 alkoxyamine onto **1** to produce PEO-MAMASG1 (**2**) and the polymerization of styrene using **2** as an initiator. PEO-acrylate (**1**) was obtained by esterification of PEO-Me ($M_n = 350 \text{ g mol}^{-1}$) with an acryloyl chloride. Addition of the MAMA-SG1 alkoxyamine to the PEO-acrylate (**1**) occurs via a thermal homolysis of the C-ON bond leading to the formation of the two radicals: the transitory alkyl radical MAMA and the stable SG1 nitroxide. The presence of the activated olefin **1** leads to an intermolecular radical 1,2-addition. The PEO-MAMASG1 (**2**) was then used to initiate the bulk polymerization of styrene at 105°C for 150 min under an inert atmosphere. The polymerization was stopped by quenching the reaction in an ice bath. The SG1-capped PEO-*b*-PS diblock copolymer (**3**) was precipitated in cold *n*-hexane, rinsed with diethyl ether and dried under vacuum at room temperature. The reaction yield of each synthesis step was determined by ^1H NMR. The final copolymer composition was determined from the ratio between the integrals of the PS phenyl ^1H signals and the methyl ^1H signal of the α -end group. The ^1H NMR compositional analysis indicated a PS block molar mass of 300 g mol^{-1} .

Liquid chromatography at critical conditions

LCCC experiments were performed on an Agilent 1100 series system (Agilent Technologies, Palo Alto, CA, USA). The chromatographic column (octadecyl-silica Equisil, $150 \times 4 \text{ mm}$, pore diameter: $100 \text{ \AA}$, particle size: $5 \mu\text{m}$), was purchased from C.I.L. Cluzeau (Sainte Foy La Grande, France) and was maintained at a constant temperature of 25°C . For all experiments, the injection volume was $20 \mu\text{L}$ and the flow

rate was set to 1 mL min^{-1} . Prior to entering the ESI source, the chromatographic effluent was split down to $50 \mu\text{L min}^{-1}$ using a zero-dead-volume tee connector.

Mass spectrometry

Detection was performed with a QStar Elite mass spectrometer (Applied Biosystems SCIEX, Concord, ON, Canada) equipped with an ESI source operated in the positive ion mode. The capillary voltage was set at 5500 V and the cone voltage at 80 V. In this hybrid instrument, ions were measured using an orthogonal acceleration time-of-flight (TOF) mass analyzer. A quadrupole mass analyzer was used for the selection of precursor ions to be submitted to collision-induced dissociation (CID) in MS/MS experiments. Nitrogen gas was used as the nebulizing gas (20 psi), the curtain gas (20 psi) as well as the collision gas. Analyst software version 2.1 (Applied Biosystems SCIEX) was used for instrument control, data acquisition and data processing for both chromatography and mass spectrometry.

RESULTS AND DISCUSSION

Determination of PEG critical conditions

The principle of using LCCC to characterize diblock copolymers is to make one block chromatographically 'invisible' (i.e. elution at the critical conditions for this block) and to separate the diblock copolymer molecules according to the size of the second block. The use of critical conditions appropriate to PEO was preferred since this approach would allow any residual PEO homopolymers (species 1 or 2) to be distinguished from copolymer molecules. Due to the amphiphilic nature of PEO-*b*-PS copolymers, the choice of a chromatographic system is difficult. On one hand, an adsorbent with a non-polar surface would lead to irreversible adsorption of large PS block molecules. On the other hand, using a polar adsorbent would allow co-oligomers to be distinguished according to the exclusion mode, thus limiting elution volume to the void volume of the column. In addition, the binary mixture should solubilize both PEO and PS blocks and should thus be carefully chosen. A non-polar adsorbent was selected as the stationary phase, as the small size of the targeted copolymer should avoid irreversible PS adsorption. The mobile phase was a binary mixture of MeOH/water containing a lithium salt. Since the aim was to fragment copolymer adducts, lithium was selected as the cationizing agent as it has been shown to provide more relevant information in MS/MS of PEO-based polymers.^{39,51–53} Moreover, successful ionization of this PEO-*b*-PS as lithiated adducts has previously been described.⁵⁰

The critical composition of the MeOH/water eluent was determined for hydroxyl-terminated PEG standards, using a 1 mM LiCl concentration. The elution volumes of small PEG standards ($200 \text{ g} \cdot \text{mol}^{-1} < M_w < 2000 \text{ g} \cdot \text{mol}^{-1}$) are reported in Fig. 1 for three tested experimental conditions. For the selected system, critical conditions were obtained with a ratio 88.8:11.2 (% v/v) MeOH/water eluent, as indicated by the same elution volume ($V_e = 2.33 \text{ mL}$) measured for all PEG chains, independently of their molecular mass. As expected for this non-polar stationary phase, enrichment of the eluent with MeOH induced a decrease in the PEG elution volume.

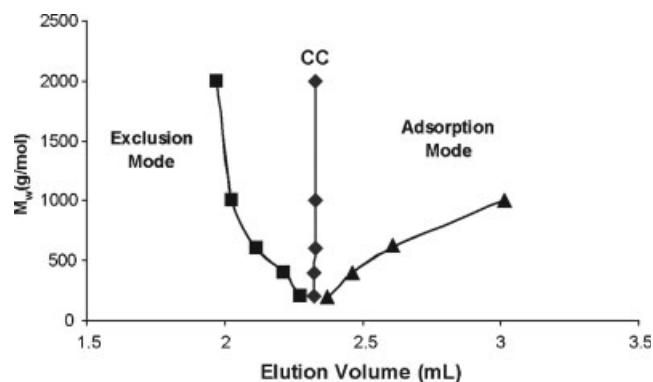


Figure 1. Elution volume of poly(ethylene glycol) standards, ranging from 200 to $2000 \text{ g} \cdot \text{mol}^{-1}$, in binary mixture of MeOH-water containing LiCl (1 mM) of different composition (% v/v): 97:3 (■); 88.8:11.2 (◆); 70:30 (▲). Adsorbent: octadecyl-silica, $T = 25^\circ\text{C}$, flow-rate: 1 mL/min , detection ESI-MS. Data result from a set of four injections ($20 \mu\text{L}$).

As shown in Fig. 1, a 97% MeOH mobile phase caused the PEG molecules to be separated according to the exclusion mode (i.e., elution volumes decrease with oligomer size), in which interactions between the mobile and the stationary phases are strong and active sites on the stationary phase are less available for polymer molecules. Conversely, with eluents containing less than 88.8% of MeOH, e.g. 70% MeOH in Fig. 1, elution volumes increased with PEG size in a classical adsorption mode.

First established for the 200–2000 Da mass range as required for the purpose of our study, these optimized critical conditions have been tested for larger size polymers. Elution volumes of PEGs with $M_w = 4$ to 10 kDa were shown to slightly increase with polymer size when they were subjected to LCCC conditions established with chains of smaller size (Fig. 2). This behaviour indicates adsorption

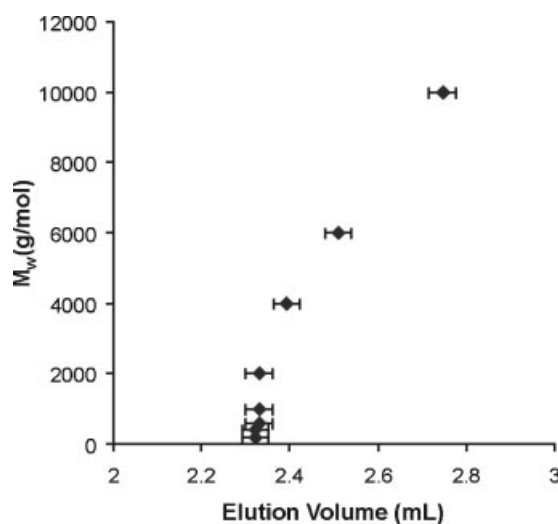


Figure 2. Elution volume of poly(ethylene glycol) standards, ranging from 200 to $10000 \text{ g} \cdot \text{mol}^{-1}$, in a 88.8:11.2 (% v/v) MeOH-water containing LiCl (1 mM) eluent. Adsorbent: octadecyl-silica, $T = 25^\circ\text{C}$, flow-rate: 1 mL/min , detection ESI-MS. Data result from a set of four injections ($20 \mu\text{L}$). Errors are expressed as $\pm \sigma$.

mode for the longer chains, for which interactions with the stationary phase are more likely. This result might suggest that the mobile phase composition does not match the critical adsorption point exactly and this effect would be more emphasized at higher molar mass. Another explanation would be that the pore diameter of the column might be too small for the higher size PEGs, which would not allow critical conditions. Alternatively, the observed chromatographic behaviour of larger PEGs might be due to the presence of a salt in the mobile phase. Critical conditions are usually optimized using pure solvents. Even in LCCC studies involving MS detection mode, salt required to promote oligomer ionization is systematically added after LCCC separation, to the collected fractions^{34,35,37,38} or post-column using a mixing Tee.³⁶ Here, a mobile phase containing the electrospray cationizing agent was used to avoid dilution effects produced by post-column addition. The effects of salts in the mobile phase on the critical adsorption point of large PEGs will be explored in a forthcoming study. Nevertheless, the validity of the optimized critical conditions could have been extended up to PEG-4000 since the difference between

elution volume data obtained for this polymer and the targeted 2.33 mL was not statistically significant.

LCCC/ESI-MS of the SG1-capped PEO-*b*-PS copolymer

The retention of the SG1-capped PEO-*b*-PS copolymer was examined at the critical conditions of the PEO block. As previously discussed, a small copolymer ($M_n(\text{PEO}) = 350 \text{ g} \cdot \text{mol}^{-1}$ and $M_n(\text{PS}) = 300 \text{ g} \cdot \text{mol}^{-1}$) was selected so that (i) the size of the PEO segment lies within the valid mass range of the critical conditions and (ii) irreversible adsorption of the PS block on the stationary phase is prevented. The LCCC/MS chromatogram obtained for the SG1-capped PEO-*b*-PS copolymer shows a total of seven peaks (Fig. 3). MS data extracted from the first peak (not shown), at a 2.27 mL elution volume, indicated the presence of residual PEO-acrylate (**1**) in the copolymer sample. The slightly lower elution volume of this PEO-acrylate, compared with the PEG 2.33 mL value, was at first intriguing since the strength of interactions with the adsorbent is expected to increase as the polarity of the end-groups decreases, as reported in studies

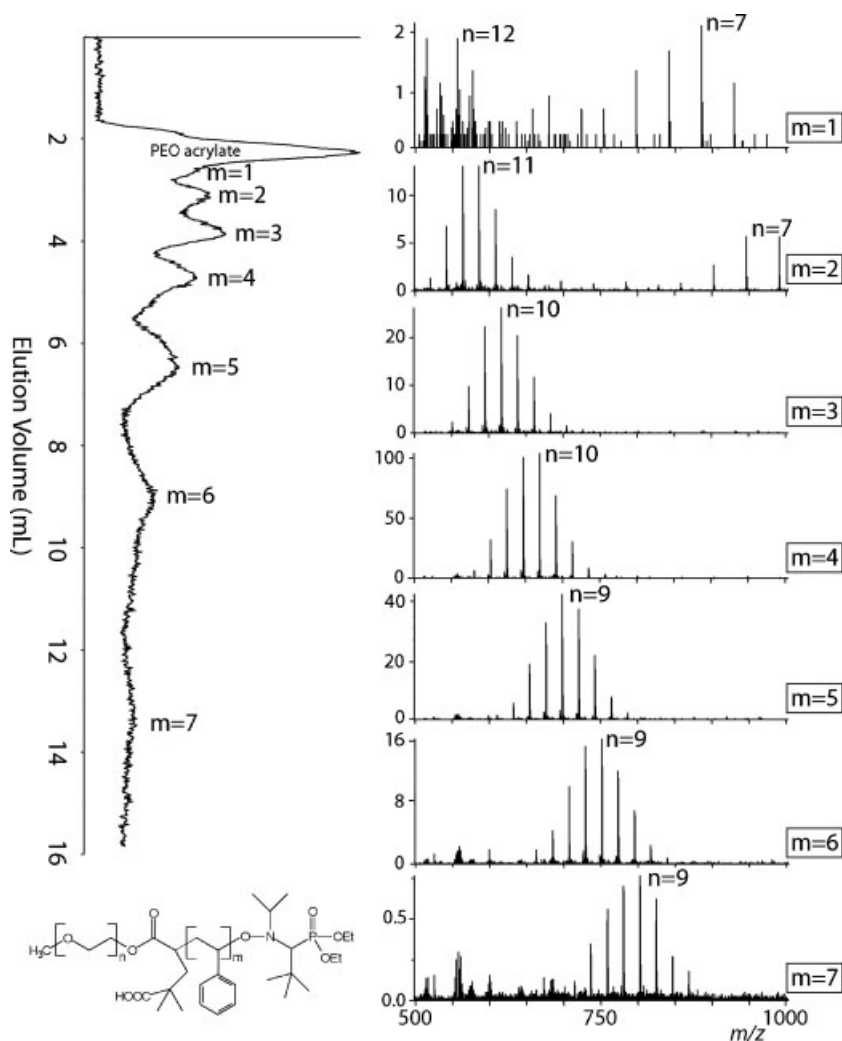


Figure 3. Chromatogram of the SG1-capped PEO_n-*b*-PS_m copolymer and MS spectra extracted from each chromatographic peak at critical conditions of PEO. Mobile phase: 88.8:11.2 (% v/v) MeOH-water mixture containing LiCl (1 mM). Adsorbent: octadecyl-silica, T = 25 °C, flow-rate: 1 mL/min.

of functional polymers by LCCC.^{10–12,19,30} However, repeatability experiments indicate that the elution volume of this PEO-acrylate (mean $V_e = 2.31$ mL, $\sigma = 0.03$, number of injections = 4) was not statistically different from the elution volume of the tested PEG standards. This result shows that the chosen chromatographic system would not be very efficient for the separation of PEG molecules with only slight structural differences. However, the specificity of the MS detection mode is shown here to circumvent this drawback.

Apart from the first PEO homopolymer peak, each peak in the chromatogram arises from copolymer molecules with different PS block size. MS data extracted from these peaks show unique PEO-like distributions of doubly lithiated molecules, as indicated by the difference of 22 m/z units measured between the ions, and the ion isotopic pattern (Fig. 3). The chromatographic peak corresponding to molecules containing only one styrene unit is poorly resolved and is actually observed as a shoulder of the PEO-acrylate peak. For the smallest copolymer molecules, i.e. when $m = 1$ –3, singly charged co-oligomers were also detected. However, as m increases, the abundance of these ions rapidly decreases compared with their doubly charged homologues. Moreover, as shown in Fig. 3, where the most abundant peak in each distribution is annotated with the corresponding number of PEO units, singly charged co-oligomer distributions were systematically detected with a bias towards low-mass oligomers compared with doubly charged species. The presence of multiply charged species produced in ESI-MS is known to facilitate detection of higher mass compounds. MS data obtained in this coupling are significantly different from spectra acquired after infusion of the same copolymer in the ionization source, where singly charged lithiated adducts were the main detected species.⁵⁰ Since the same cone voltage was applied in both cases, the observed decrease of co-oligomer charge state in the infusion experiments would arise from the simultaneous presence of residual PEO-acrylate and co-polymer molecules in the ionization source. The homopolymer is expected to have a

higher cation affinity than the copolymer, and would readily capture lithium through charge exchange reactions occurring in the ESI source.⁵⁴ The PEO block being chromatographically 'invisible', copolymer molecules with a given PS block size are eluted in the same volume, regardless of the PEO block length. The elution volume of co-oligomers increases with m , the number of styrene units, accounting for the increase of interactions between co-oligomers and the stationary phase as molecule hydrophobicity increases. As the masses of both the end and the block linkage-groups are known, the monomer composition of each ion could be readily calculated from MS data. The m/z values of doubly lithiated co-oligomers observed in Fig. 3 are reported in Table 1 (using $m/z = (44n + 104m + 467 + 2 \cdot 7)/2$), where n and m are the number of ethylene oxide and styrene monomers, respectively. These data are consistent with the indicated polymerization degree of the two blocks, as determined by ^1H NMR. Compared with data obtained for singly lithiated molecules in the infusion mode,⁵⁰ a larger number of co-oligomers was detected as doubly charged adducts when the number of styrene units increases above $m = 4$. Moreover, the distributions of co-oligomers in the +2 charge state were slightly shifted to higher n values. The higher yield observed for doubly charged co-oligomers as the PS block size increases could result from attachment of one Li^+ cation to the phenyl rings since aromatic groups were reported to bind alkali metal ions quite strongly.^{55,56} In contrast, copolymer molecules with eight styrene units were not observed in the chromatogram while they were detected in mass spectra acquired in the infusion mode.⁵⁰ These molecules might be too hydrophobic to be eluted from the column using the selected mobile phase. However, it should be emphasized that only two of these co-oligomer molecules (i.e., PEO₇₋₈-*b*-PS₈) were observed in the infusion mode and they were detected with a very low abundance. Increase of the LCCC peak width as m increases might well account for the observed decreasing signal-to-noise ratios in MS spectra and would result in a poor detection of co-oligomers with

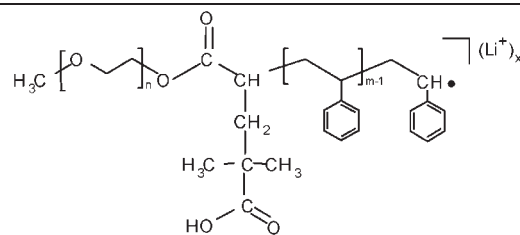
Table 1. Detected m/z values and elution volumes (V_e) of doubly lithiated SG1-capped PEO-*b*-PS co-oligomers. n and m indicate the number of ethylene oxide and styrene units, respectively

		m						
		1	2	3	4	5	6	7
V _e (mL)		2.62	3.15	3.87	4.71	6.45	8.95	13.32
n	4					588.8	640.8	
	5				558.8	610.8	662.8	714.9
	6			528.7	580.8	632.8	684.8	736.9
	7		498.7	550.8	602.8	654.8	706.8	758.9
	8		520.7	572.8	624.8	676.8	728.8	780.9
	9		542.7	594.8	646.8	698.8	750.9	802.9
	10	512.8	564.8	616.8	668.8	720.8	772.9	824.9
	11	534.8	586.8	638.8	690.8	742.8	794.9	846.9
	12	556.8	608.8	660.8	712.8	764.9	816.9	868.9
	13	578.8	630.8	682.8	734.8	786.9	838.9	890.9
	14	600.8	652.8	704.8	756.9	808.9	860.9	
	15		674.8	726.8	778.9	830.9		
	16			748.8	800.9			
	17			770.8				

Tandem mass spectrometry of LCCC separated species

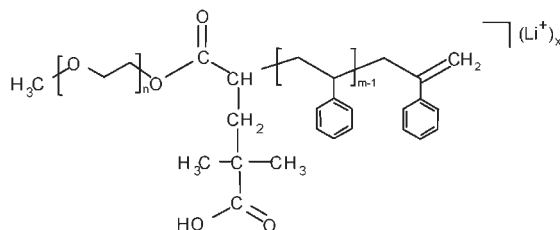
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Structural information on the SG1 end-group could be reached from weak abundance product ions detected in the low m/z range of the MS/MS spectrum. The proposed mechanisms were supported by accurate mass measurements and Table 2 summarizes the structures of fragment

Table 2. Structure of product ions observed in co-oligomer MS/MS spectra. The superscript *x* indicates that some ions were observed in both +1 and +2 charge states $\mathbf{b}_{n,m}^{x+}$

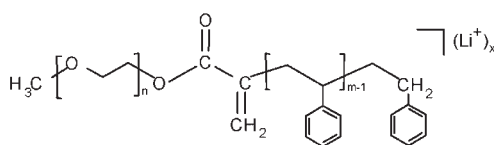
$$m/z = 44n + 104m + 180 \text{ for } x = 1$$

$$m/z = 22n + 52m + 93.5 \text{ for } x = 2$$

 $\mathbf{\beta}_{n,m}^{x+}$

$$m/z = 44n + 104m + 193 \text{ for } x = 1$$

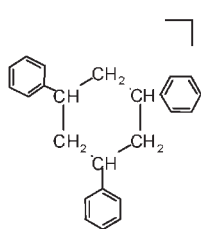
$$m/z = 22n + 52m + 100 \text{ for } x = 2$$



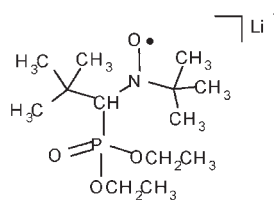
$$m/z = 44n + 104m + 93 \text{ for } x = 1$$

 $\mathbf{[b_{n,m}-MAMA]^{x+}}$

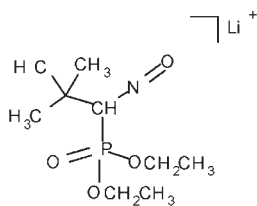
$$m/z = 22n + 52m + 50 \text{ for } x = 2$$



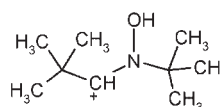
$$m/z \text{ 319.1}$$

 $\mathbf{[SG1^{+}Li]^{+}}$

$$m/z \text{ 301.1}$$



$$m/z \text{ 244.1}$$



$$m/z \text{ 158.1}$$

ions observed in co-oligomer MS/MS spectra. As already reported for singly lithiated co-oligomers,⁵⁰ it should be noted that specific dissociation of the PEO block was not observed to occur in MS/MS of doubly charged molecules, suggesting (i) these rearrangement reactions do not compete effectively with dissociations of the odd-electron product ions and/or (ii) the C–C and C–O bonds in the PEO chain are stronger than the C–C(Ph) bonds in the PS chain.

CONCLUSIONS

On-line coupling of LCCC with MS has been successfully applied for the structural characterization of a PEO-*b*-PS copolymer. LCCC experimental conditions have been optimized using a mobile phase which contains the electrospray cationizing agent, avoiding dilution effects commonly observed while performing post-column salt addition. Liquid chromatography at the PEO critical conditions allows the separation of co-oligomer molecules according to the styrene block size, thus providing greatly simplified mass spectra from each separated species. In MS mode, the size of repeat units, as well as the end-group masses and degree of polymerization, could be reached in a straightforward data analysis. Implementation of the coupling in the MS/MS detection mode further allows unambiguous determination of the block size, taking advantage of specific dissociation pathways allowed by the labile SG1 end-group, and the microstructure of the copolymer could be fully characterized. Compared with results obtained in infusion mode, higher charge state molecules could be produced and were shown to dissociate according to the same fragmentation pathways previously established for singly charged co-oligomers. In contrast, co-oligomers with a large PS block, poorly detected in the infusion mode, were no longer observed using the LCCC/MS coupling, probably due to a decrease in the ionization yield of these molecules.

Acknowledgements

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